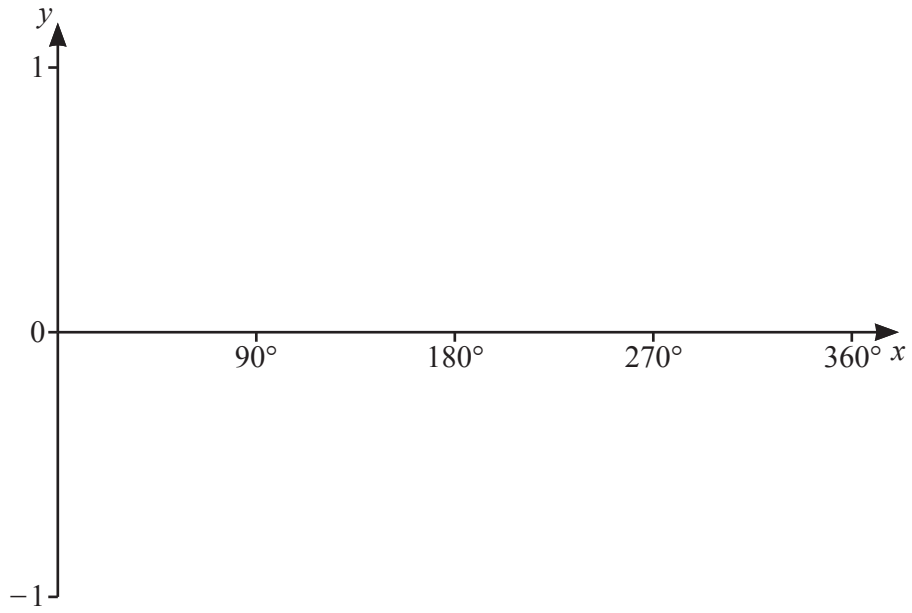
0580_s20_qp_41_Q8

8 **(a)** **(i)** On the axes, sketch the graph of $y = \sin x$ for $0° \leqslant x \leqslant 360°$.



[2]

(ii) Describe fully the symmetry of the graph of $y = \sin x$ for $0° \leqslant x \leqslant 360°$.

.....

..... [2]

(b) Solve $4\sin x - 1 = 2$ for $0° \leqslant x \leqslant 360°$.

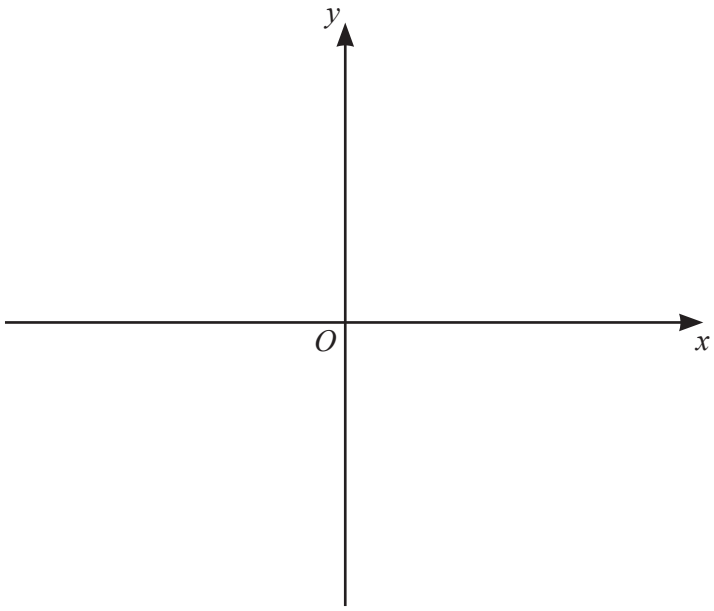
$x =$ ..... and $x =$ ..... [3]

0580_s20_qp_41_Q8

(c) **(i)** Write $x^2 + 10x + 14$ in the form $(x + a)^2 + b$.

..... [2]

(ii) On the axes, sketch the graph of $y = x^2 + 10x + 14$, indicating the coordinates of the turning point.



[3]